

MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.5 Parametric Equations of Lines

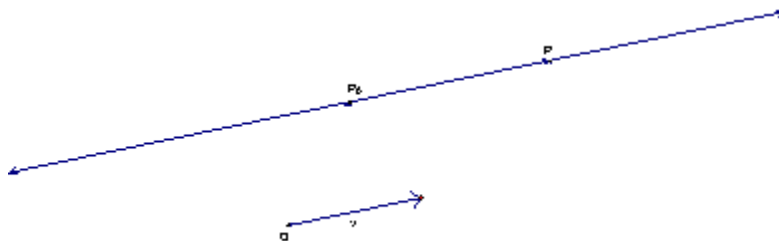
Question: In Algebra 1 what information do you need in order to find the equation of a line?

Motivation: What information do we need to determine the equation of a line in 3-dimensions and higher?

We can think of identifying a line in 3-space if we know:

Definition: If a line, l , is parallel to a vector $\mathbf{v}$, then we call $\mathbf{v}$ a direction vector for l .

In the picture we can see, O is the origin, P is an arbitrary point on the line, and $\mathbf{v}$ is our direction vector.



We then develop the following terminology.

Terminology: The vector equation of a line L is $\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$ where $\mathbf{r}_0$ is the position vector of a point on the line and $\mathbf{v}$ is a vector parallel to the line. Each real number value of t (the parameter) gives the position vector of a point on L .

Note: This idea of a vector equation for a line works for two dimensions and higher.

Terminology: In 3 dimensions if we let $\mathbf{r} = \langle x, y, z \rangle$, $\mathbf{r}_0 = \langle x_0, y_0, z_0 \rangle$, and $\mathbf{v} = \langle a, b, c \rangle$, the vector equation of the line L becomes: $\langle x, y, z \rangle = \langle x_0, y_0, z_0 \rangle + t\langle a, b, c \rangle \Rightarrow \langle x, y, z \rangle = \langle x_0 + at, y_0 + bt, z_0 + ct \rangle$.

This leads to other ways of representing the equation of a line. For example, we could represent the line L as three separate equations by equating components of the vectors on each side of the vector equation.

Example 1: Find the vector equation and parametric equations of the line that passes through the point $(2, -3, 4)$ and is parallel to the vector $-\mathbf{i} + \mathbf{j} + \mathbf{k}$.

Example 2:

a) Find parametric equations for the line that passes through the points $(2, -3, 4)$ and $(4, 2, 1)$.

b) Find the point at which the line that you found in (a) intersects the xz -plane.

c) What restrictions should you place on t if you want your parametric equations from (a) to determine only the line segment connecting $(2, -3, 4)$ and $(4, 2, 1)$?

Remark: If two lines have parallel direction vectors, the two lines are parallel. If two lines intersect at a point we say that the two lines intersect. If two lines do not intersect and are not parallel we say that the lines are skew.

Example 3: Suppose the line L_1 is given by: $\langle x, y, z \rangle = \langle -3, 0, 1 \rangle + t \langle 1, 2, 1 \rangle$, and the line L_2 is given by: $\langle x, y, z \rangle = \langle 1, 4, -7 \rangle + s \langle -1, -1, 2 \rangle$. Determine if these lines are parallel, intersecting, or skew.

Example 4: Suppose the line L is given by: $\langle x, y, z \rangle = \langle -3, 0, 1 \rangle + t \langle 1, 2, 1 \rangle$. Find the exact distance between this line and the point P which has coordinates $(1, 1, 1)$.